EE4002: Introduction to Image Processing

HW2

Deadline: May 23, 2025

Question 1 (15 Points)

A periodic time function $\tilde{x}[n]$ has period 8. The function and its Fourier series coefficients are as follows:

$$\begin{aligned}
\tilde{x}[0] &= 2 \quad \tilde{X}[0] = 3 \\
\tilde{x}[1] &= 3 \quad \tilde{X}[1] = 8.07 \\
\tilde{x}[2] &= 1 \quad \tilde{X}[2] = 5 \\
\tilde{x}[3] &= 2 \quad \tilde{X}[3] = 6.07 \\
\tilde{x}[4] &= 1 \quad \tilde{X}[4] = 1 \\
\tilde{x}[5] &= 2 \quad \tilde{X}[5] = 6.07 \\
\tilde{x}[6] &= 1 \quad \tilde{X}[6] = 5 \\
\tilde{x}[7] &= 3 \quad \tilde{X}[7] = 8.07
\end{aligned}$$

Sketch and dimension $\tilde{X}(e^{j\omega})$, the DTFT of $\tilde{x}[n]$.

Question 2 (35 Points)

Figure 1 above shows the input $x[n]$ and unit sample response $h[n]$ of a simple discrete-time system. Note that $x[n]$ is nonzero only for $0 < n \leq 39$. Although $h[n]$ is defined for $0 \leq n \leq 7$, this function is nonzero only for $2 \leq n \leq 5$. The specific numerical values of $x[n]$ and $h[n]$ are important for the purposes of this question.

(a) Consider first $y[n]$, the result of the linear convolution of $x[n]$ and $h[n]$. For exactly which values of n will $y[n]$ be nonzero?

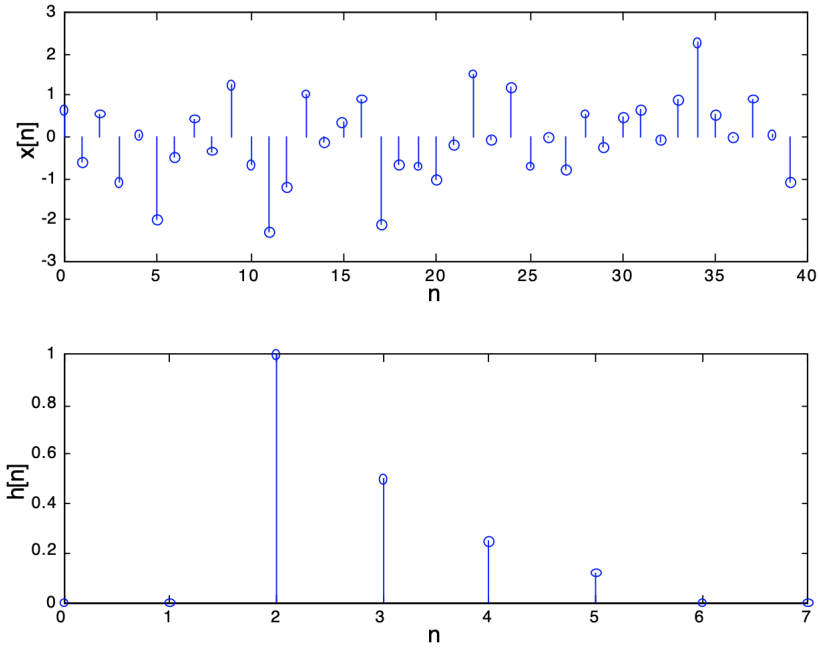


Figure 1: Input and unit sample response of a discrete-time filter

(b) Now suppose that we compute $X[k]$ and $H[k]$, the 40-point DFTs of $x[n]$ and $h[n]$ (The 40-point DFT of $h[n]$ is obtained by appending an appropriate number of zeros to the original time function). Let $y_2[n]$ be the inverse 40-point DFT of the product of $X[k]$ and $H[k]$. For which values of n will $y_2[n]$ be equal to $y[n]$?

(c) We would now like to obtain the linear convolution of $x[n]$ and $h[n]$ using an implementation of the overlap-add method. Our implementation includes the following steps:

- Segment $x[n]$ into two or more abutting sections of length 20.

- Extract the first 6 points of $h[n]$ (which include all of the nonzero points of the function)

As usual with the overlap-add method, sequences of output samples are generated by obtaining the circular convolution of $h[n]$ with the abutting sections of $x[n]$, and then adding the results obtained from these convolutions together.

1. What is the minimum length of the DFT needed in order to obtain the circular convolution of $h[n]$ with the sections of $x[n]$ without having the results corrupted by temporal aliasing? Assume that zeros are appended to both $h[n]$ and the abutting sections of $x[n]$ before computing their DFTs.

2. How many sections of $x[n]$ will be needed to obtain a complete result that is equal to $y[n]$?

3. How are the output samples combined? Be specific, stating the manipulations of inputs and outputs in terms of the values of the time index n.

Question 3 (50 Points)

(a) A causal LSI system has zeros at $z = 0$, $z = 1$, and $z = 2$, and its poles are located at $z = -1$, $z = -1 + j$, and $z = -1 - j$. It is described by the difference equation

$$y[n] = 3y[n-1] - 4y[n-2] - 2y[n-3] + x[n] - 3x[n-1] + 2x[n-2] \tag{1}$$

1. Draw a signal flow diagram that describes the direct form II implementation of the filter including numerical values for all constants.

2. Draw a signal flow diagram that describes the transposed direct form II implementation of the filter including numerical values for all constants.

3. Draw a signal flow diagram that describes the cascade form implementation of the filter including numerical values for all constants. The filter should be implemented in second-order sections (i.e. sections with up to two poles and two zeros) with real coefficients.

(b) A second causal LSI system is represented by the transfer function

$$H(z) = \frac{z(z-1)}{(z+1)(z-2))} \tag{2}$$

Draw a signal flow diagram that describes the parallel form implementation of the filter including numerical values for all constants. Although the parallel-form implementation is usually performed using second-order sections (as in the case of the cascade form), please solve this part of the problem with an implementation that uses first-order sections, each representing up to one pole and one zero.

(c) Now consider the FIR filter that is represented by the signal flowgraph below Draw a signal low diagram that describes the direct form implementation of the filter including

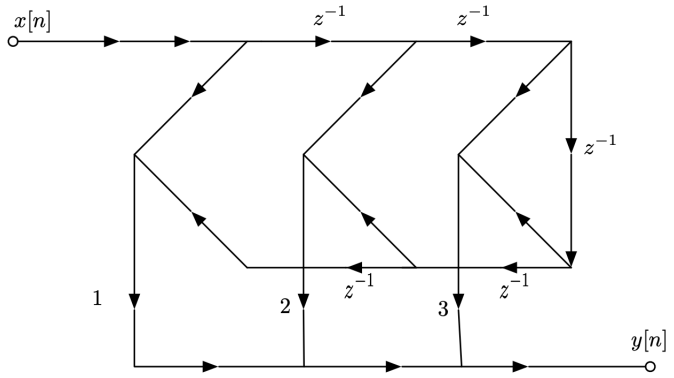


Figure 2: Illustration of signal flowgraph related to Q3

numerical values for all constants.